

PROBLEMS

Question 1 – Problem Statement

A hospital needs to assign **three doctors (D1, D2, and D3)** to **three shifts (Morning, Afternoon, and Night)**. The assignment must satisfy the following constraints:

- D1 cannot work the Night shift.
- D2 must work before D3 (Morning < Afternoon < Night).
- Only one doctor can be assigned to each shift.
- D3 cannot work the Morning shift.
- Every doctor must be assigned exactly one shift.

Use the **Backtracking Search algorithm** to find a valid schedule by showing all intermediate assignments, constraint checks, and the final valid assignment.

Question 2 – Problem Statement

A robot operates in a **5 × 5 grid environment** and must travel from the **Start (S)** position to the **Goal (G)** while avoiding obstacles. The robot can move only in the four cardinal directions: **Up, Down, Left, and Right**. Each movement has a cost of **1**, and the robot cannot move through blocked cells. The **Manhattan Distance heuristic** is used to estimate the distance to the goal.

Apply the **Breadth-First Search (BFS)** algorithm to:

- Show the step-by-step node expansion.
 - Display the queue updates after each expansion.
 - Determine the optimal path.
 - Calculate the total path cost.
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Question 3 – Problem Statement

An autonomous rescue robot is deployed in a **disaster-affected building** to locate and rescue survivors. The robot starts at position **S** and must reach the survivor located at **G**. The building contains blocked paths, risky zones, and limited visibility.

The robot must satisfy the following constraints:

- It can move only **Up, Down, Left, and Right**.
- It cannot move through blocked cells.

- It can observe only adjacent cells (partially observable environment).
- Each normal movement costs **1**, while entering a risky zone adds an extra cost of **2**.
- The robot must minimize the total travel cost while ensuring safe navigation.
- It must continuously update its knowledge as new information is discovered (online search).

Analyze the problem by:

1. Explaining how each constraint affects the robot's decision-making.
2. Developing a suitable solution using an appropriate search strategy (such as **Uniform Cost Search** or an **Online Search Agent**).
3. Explaining the AI concepts involved, including **intelligent agents**, **rationality**, and the **environment type**.
4. Justifying why the chosen approach is the most appropriate under the given constraints.